

Assignment 1 - Patient Clinical Data Analysis

Questions are followed by answers. Clinical flags below are analytical screening assumptions, not medical diagnoses.

Q1. Load the dataset into a Pandas DataFrame.

Answer:

```
import pandas as pd
df = pd.read_csv("patient_clinical_data_raw.csv")
```

Q2. Display the first 10 and last 10 records.

Answer:

```
print(df.head(10))
print(df.tail(10))
```

Q3. Find the number of rows and columns.

Answer:

Raw data: **10,100 rows x 14 columns**. After removing 100 exact duplicate rows, the working dataset has 10,000 rows.

Q4. Display the column names and their data types.

Answer:

Columns: Patient_ID (object), Age (float), Gender (object), Department (object), Blood_Pressure (object), Heart_Rate (float), Temperature (float), Cholesterol (float), Glucose (float), Diagnosis (object), Admission_Type (object), Length_of_Stay (integer), Treatment_Cost (float), Readmission (object). Use df.dtypes to display them.

Q5. Generate a statistical summary of the numerical columns.

Answer:

Use df.describe(). Selected raw-data mean / min / max: Age 47.42 / -5 / 150; Heart_Rate 77.93 / 45 / 122; Temperature 98.40 / 95 / 102.9; Cholesterol 195.19 / 100 / 326; Glucose 106.20 / 60 / 209; Length_of_Stay 5.03 / 1 / 14; Treatment_Cost 100,538.42 / 13,492.02 / 315,212.62.

Q6. Identify the number of unique values in each column.

Answer:

Patient_ID 10,000; Age 92; Gender 5; Department 9; Blood_Pressure 3,645; Heart_Rate 75; Temperature 76; Cholesterol 206; Glucose 145; Diagnosis 8; Admission_Type 3; Length_of_Stay 14; Treatment_Cost 9,998; Readmission 2.

Q7. Identify all columns containing missing values.

Answer:

Age 302; Heart_Rate 302; Temperature 304; Cholesterol 303; Glucose 302. All other columns have no missing values.

Q8. Calculate the percentage of missing values in each column.

Answer:

Age 2.99%; Heart_Rate 2.99%; Temperature 3.01%; Cholesterol 3.00%; Glucose 2.99%. All other columns: 0%.

Q9. Handle missing values in the numerical columns appropriately.

Answer:

For analysis, replace missing numerical measurements with their column medians. For Age, first treat impossible ages as missing, then use the median of valid ages (47 years). Preserve a flag for imputed values in production.

Q10. Identify and remove duplicate records.

Answer:

df.duplicated().sum() returns **100**. df.drop_duplicates() leaves **10,000** unique records.

Q11. Check the Age column for invalid values.

Answer:

Valid ages were defined as 0 through 120. There are **15 invalid records**: ten ages of -5 and five ages of 150.

Assignment 1 - Patient Analysis (Q12-Q22)

Q12. Correct or handle patients whose age is outside a reasonable range.

Answer: Set ages outside [0, 120] to missing, then impute missing/invalid ages with the median valid age (47). After deduplication, 300 ages remain missing and 15 are invalid: **315 rows** need imputation.

Q13. Identify inconsistent categorical values such as Gender and Department.

Answer: Gender has five raw spellings: Male, Female, male, MALE, and padded FEMALE. Department has nine raw spellings, including lower-case and whitespace variants of Cardiology and Neurology.

Q14. Standardize the categorical values.

Answer: Strip whitespace and title-case values (plus an explicit alias map if required). This yields Gender={Female, Male} and six departments: Cardiology, General Medicine, Neurology, Oncology, Orthopedics, Pediatrics.

Q15. Create Age_Group: <18 Child; 18-40 Young Adult; 41-60 Middle Age; >60 Senior.

Answer: Use left-closed bins [0,18), [18,41), [41,61), [61,121). Counts after cleaning/imputation: Child 479; Young Adult 2,901; Middle Age 4,377; Senior 2,243.

Q16. Separate Blood_Pressure into Systolic and Diastolic.

Answer: Split on "/" and convert both parts to numeric: Systolic=first part, Diastolic=second. All 10,000 cleaned records parse.

Q17. Create a High_Risk column based on suitable clinical conditions.

Answer: For a transparent screening flag (not diagnosis), flag age ≥ 65 , systolic ≥ 140 , diastolic ≥ 90 , glucose ≥ 126 , cholesterol ≥ 240 , heart rate > 100 , or temperature ≥ 100.4 F. Result: **6,760 records**. The glucose cutoff assumes fasting measurement.

Q18. Create Cost_Per_Day using treatment cost and length of stay.

Answer: Cost_Per_Day = Treatment_Cost / Length_of_Stay. Mean: **\$20,757.39 per patient-day**.

Q19. Calculate the average age of patients.

Answer: After deduplication and valid-age handling, mean Age = **47.39 years**.

Q20. Find the number of patients in each department.

Answer: Cardiology 1,761; Neurology 1,712; Oncology 1,662; Orthopedics 1,644; Pediatrics 1,614; General Medicine 1,607.

Q21. Calculate average treatment cost for each department.

Answer: Cardiology \$102,040.80; Neurology \$101,224.79; General Medicine \$100,711.12; Orthopedics \$100,302.54; Oncology \$100,039.38; Pediatrics \$99,036.60.

Q22. Find average length of stay for each diagnosis.

Answer: Asthma 4.99 days; Cancer 5.00; Diabetes 5.08; Fracture 5.03; Heart Disease 5.00; Hypertension 5.10; Infection 5.02; Migraine 5.05.

Q23. Identify the top 10 patients with the highest treatment cost.

Answer:

P09106 \$315,212.62; P01900 \$288,519.47; P06995 \$286,702.48; P09616 \$285,387.70; P07933 \$278,888.31; P03835 \$273,491.74; P01350 \$273,458.79; P01358 \$273,007.12; P08810 \$271,716.86; P07801 \$269,594.52.

Q24. Find the department with the highest average treatment cost.

Answer:

Cardiology, averaging \$102,040.80.

Q25. Calculate the percentage of patients who were readmitted.

Answer:

1,854 of 10,000 patients: **18.54%**.

Q26. Compare readmission rates across departments.

Answer:

Neurology 20.56%; Cardiology 19.88%; Pediatrics 18.53%; Orthopedics 18.37%; General Medicine 17.80%; Oncology 15.94%. Neurology is highest and Oncology lowest.

Q27. Find the average cholesterol and glucose level by diagnosis.

Answer:

Diagnosis: cholesterol / glucose. Asthma 197.09 / 105.99; Cancer 195.38 / 104.73; Diabetes 195.02 / 107.37; Fracture 194.13 / 105.05; Heart Disease 196.66 / 107.39; Hypertension 194.12 / 107.27; Infection 193.59 / 105.39; Migraine 195.65 / 106.19.

Q28. Identify patients with high glucose, high cholesterol, and high heart rate.

Answer:

Using glucose ≥ 126 , total cholesterol ≥ 240 , and heart rate > 100 : **2,483** high-glucose records; **973** high-cholesterol records; **295** high-heart-rate records. These flags are screening criteria only.